

# Server-side measurement study — what is in place

## 1. Probes built (no edits to llama.cpp)

- entropy\_probe — per-step Shannon H, top-1, top-5 (LSE-stable)
- attention\_probe — disables flash-attn; cb\_eval grabs every kv-software layer
- Both built standalone via examples/, link libllama.so + libggml.so

## 2. Workload = LongBench (Bai et al., ACL 2024)

- 4 tasks × 6 prompts = 24: qasper · multifactqa\_en · triviaqa · samsum
- Same token family used by KVSwap, KIVI, CAKE, SnapKV
- Contexts truncated to ~2 k chars to fit 4 096-token ctx

## 3. Study run (Phase D)

- Llama-3.2-1B-Instruct Q4\_K\_M, GPU, seed 42, max\_tokens 64
- 994 decode steps written to logs/study\_full.csv
- Per-prompt attention sidecars (.attn.bin) — 24 files, full source x 16 layers

## 4. Offline analysis (Phase E.1)

- Entropy distribution by task (Kruskal-Wallis  $p = 0.19$  → task-agnostic)
- Entropy auto-correlation lag-1 = 0.27 → predictable for control
- $\text{H}_{\text{t}} \text{ top-1 } \rho = -0.298$  ( $n = 986$ ,  $p < 10^{-21}$ )

## 5. Next = Phase E.2 (rigorous validation)

- prune\_probe — re-run each decode with K oldest KV entries evicted
- Measure  $\text{KL}(\text{P}_{\text{full}} \parallel \text{P}_{\text{pruned}})$  for  $K \in \{16, 64, 256\}$
- Hypothesis test: does  $\text{H}_{\text{t}}$  correlate with  $\text{KL}_{\text{t}}$  ? → the answer that closes the safety-gate loop